

Branchline Coupler Design

Nishanth Kiruthivasan
College of Engineering
Northeastern University
Boston, MA, USA

Abstract—This project focuses on the design and simulation of a 3 GHz, 50Ω Branchline Coupler using Keysight ADS. The objective is to develop a two-metal layer PCB layout with copper as the conductive material and FR4 as the dielectric substrate with a dielectric constant of 4.6. The process begins with impedance line calculations using the Control Impedance Line Designer in ADS, followed by piecewise simulations of individual segments using S-parameter models. Subsequently, a complete EM simulation of the entire coupler is performed to validate performance. The report documents the design choices, simulation results, and rationale behind the selected methodology.

I. INTRODUCTION

Branchline Couplers, also called quadrature hybrids, are 3dB directional couplers with 90° phase difference between through and coupled ports. Typically implemented in microstrip or stripline form, they divide input power equally between output ports, ideally delivering no power to the isolated port. Figure 1 shows a basic branchline coupler design.

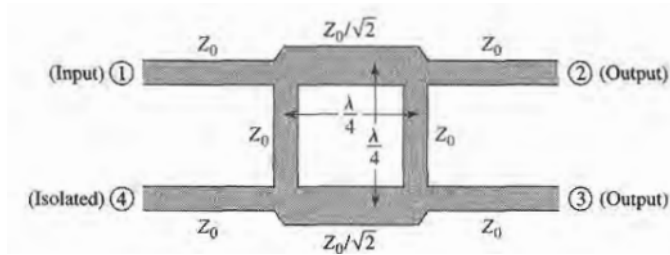


Fig. 1. A typical Branchline Coupler [1]

In an ideal configuration with matched ports, power entering port 1 splits equally between ports 2 and 3, with a 90° phase shift. Port 4 receives no power. The setup comprises transmission lines with impedances of 50Ω and 50/√2Ω. The output S-parameter matrix calculated based on even-odd mode analysis is shown in Figure 2. This matrix is symmetrical. This essentially means that $S_{21} = S_{12}$, $S_{31} = S_{13}$ and $S_{41} = S_{14}$. The expected S-parameter plot is shown in Figure 3. [1]

This project focuses on designing a 3 GHz, 50Ω Branchline Coupler using electromagnetic (EM) simulation in Keysight ADS. The design process involves calculating preliminary line dimensions using the linecalc and designing a schematic using T-line components and simulating for S-parameters. Then using the Controlled Impedance Line Designer (CILD) individual segments were simulated and the dimensions were finalized. Using these finalized dimensions for the branchline coupler was designed and an EM simulation was conducted.

$$[S] = \frac{-1}{\sqrt{2}} \begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix}$$

Fig. 2. Expected S-parameter Matrix [1]

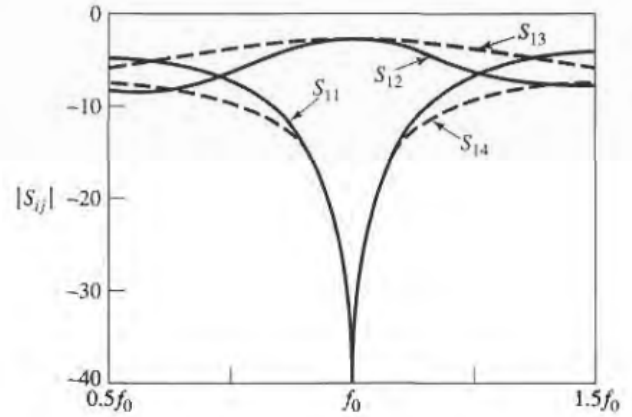


Fig. 3. Expected S-parameter plot [1]

II. DESIGN METHODOLOGY

The design methodology can be divided into 4 parts:

- Linecalc Analysis
- Schematic Simulation
- Piecewise EM Simulation
- Total EM Simulation

A. Linecalc Analysis

The process begins by configuring the MSUB component to define the substrate's characteristics, including:

- Dielectric Constant (ϵ_r) = 4.6
- Substrate Height (H) = 10 mils
- Conductor Thickness (T) = 17μm
- Dielectric Loss Tangent (TanD) = 0.02

Linecalc is then used to determine the exact length and width for a given impedance and electrical length. For the input, output and isolation transmission lines the impedance is supposed to be 50Ω. The width given by the linecalc tool was found to be 457.084 μm for the input port, through port,

coupled Port and isolation Port arms. The length was taken as 5 mm at random.

With these transmission lines ready the arms connecting these must be constructed. Off these the 35.35Ω arms must have an impedance of $50/\sqrt{2}\Omega$ and the 50Ω 90° arms need to have 50Ω . They also must have a length of $\lambda/4$. This essentially means that the electrical length of these arms are 90° . This came out to 13.2781 mm as given by linecalc.

To connect these lines T-junctions have been used with the widths of the arms matching their nearby transmission lines.

B. Schematic Simulation

With the dimensions obtained, the schematic was designed as shown in Figure 4. The schematic was terminated with ground terminations and an S-parameter block was placed to complete the simulation. The frequency was swept from 1.5 GHz to 4.5 GHz to capture the desired performance.

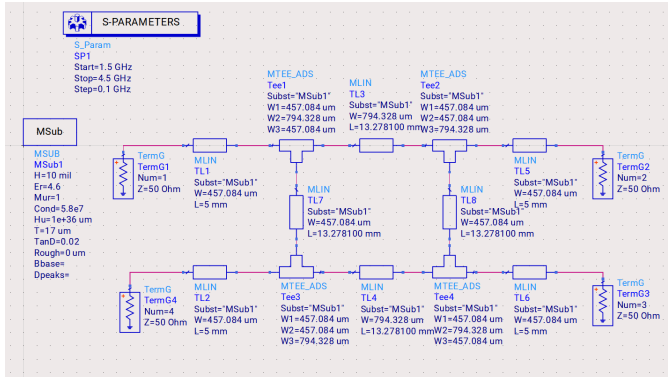


Fig. 4. Schematic

The result obtained is shown in Figure 5. The analysis of results is as follows:

- S_{11} , S_{22} , S_{33} and S_{44} are well below -25 dB, indicating low reflection.
- S_{21} and S_{31} are around -3.5 dB, indicating minimal loss.
- S_{41} is less than -40 dB, indicating high isolation.
- The phase difference between S_{21} and S_{31} is 90° .

In an ideal condition, the reflection coefficients and isolation must be less than -15 dB, insertion loss must be as near to 0 as possible and the phase difference between the signals received by through port and coupled ports must be 90° . Additionally the S-parameter matrix must be symmetrical. As seen from the results (Figure 5) this was achieved to high satisfaction.

C. Piecewise EM Simulation

Before proceeding with the complete layout and EM simulation, a substrate was created (Figure 6) and individual segments of the branchline coupler were simulated separately to ensure accurate performance. Since the initial dimensions obtained from linecalc may not be highly precise, CILD was used to validate and fine-tune the dimensions. These were implemented and piecewise simulation results were obtained. These values are more suitable in EM perspective than the ones

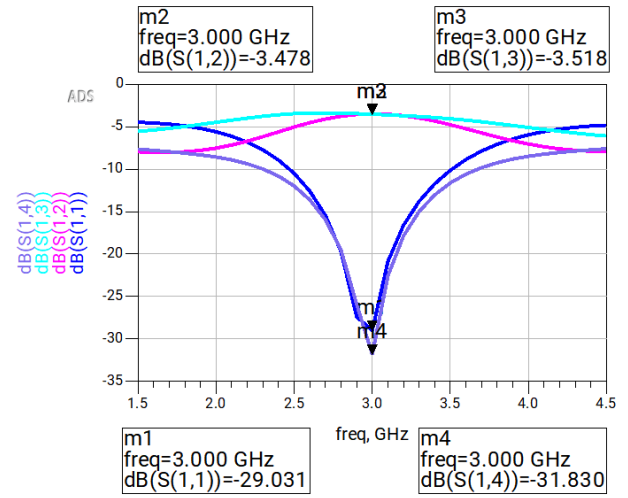


Fig. 5. Schematic Simulation Result

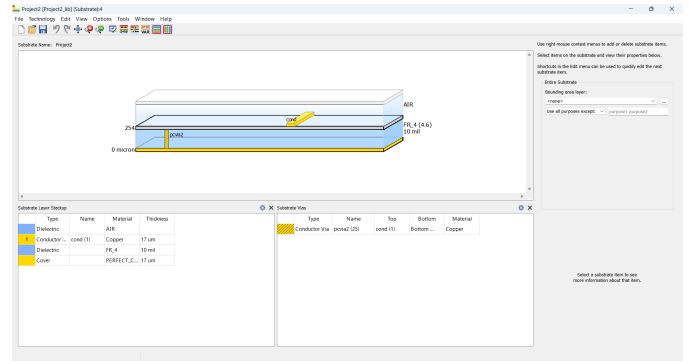


Fig. 6. Substrate

given by linecalc as these consider a lot more parameters to calculate the final values.

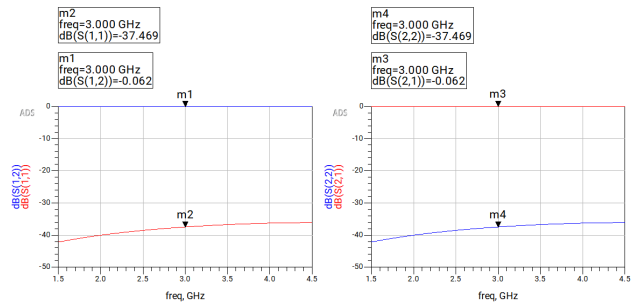


Fig. 7. 50Ω transmission line simulation result

By analyzing the simulation result of the 50Ω transmission line (Figure 7) we find that the insertion loss is nearly zero and the reflection losses are less than -35 dB, which is very good for a transmission line. This same transmission line will be used for input, through, coupling and isolation ports.

Analysis of the simulation result of the 35.35Ω transmission line (Figure 8) shows fair insertion loss but poor reflection loss.

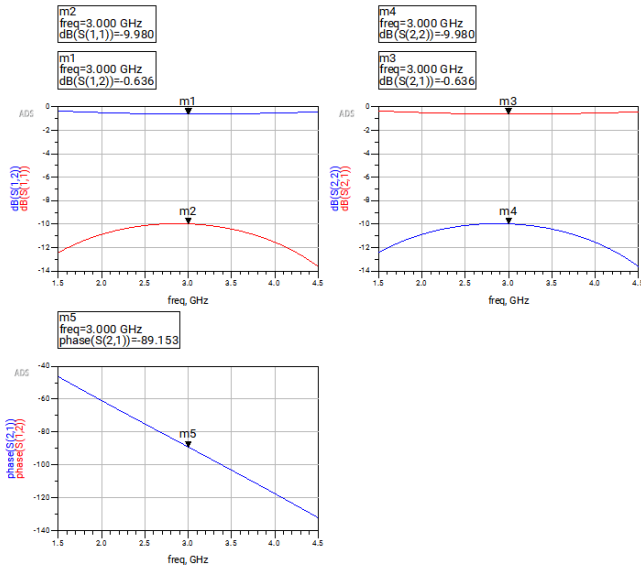


Fig. 8. 35.35Ω transmission line simulation result

The main point to note here is the phase of S21. It is shown to be almost 90°. This is because this line has length of $\lambda/4$.

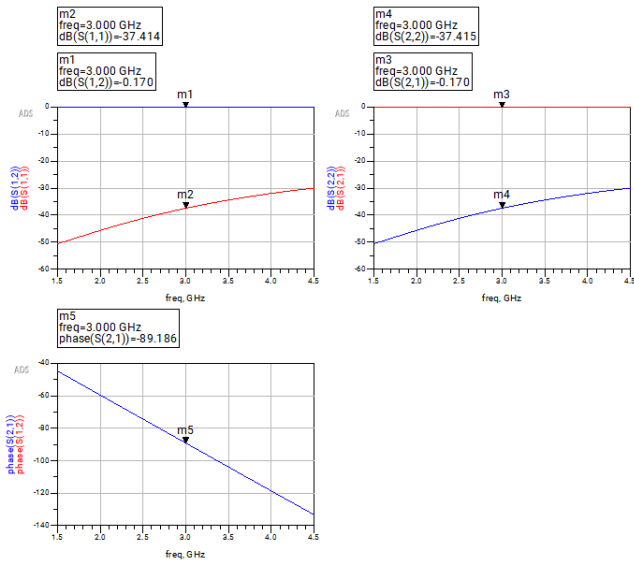


Fig. 9. 50Ω 90° transmission line simulation result

Finally, analysis of the 35.35Ω transmission line simulation result (Figure 9) outlines very low insertion loss and very good reflection loss. Similar to 35.35Ω transmission line the phase of S21 is found to be almost 90°. This is the output expected for a transmission line with a length of $\lambda/4$.

D. Total EM Simulation

Since piecewise simulations have provided the expected value, total EM simulation can be done. For this purpose the transmission lines need to be connected to one another.

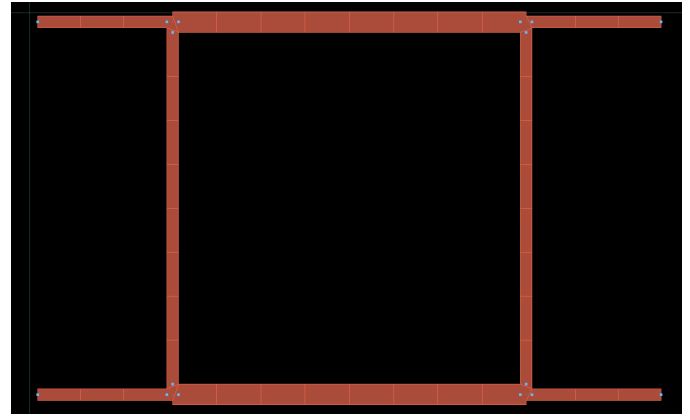


Fig. 10. Layout

This was achieved using the generate/update layout option in the schematic window. Through this the software placed and connected the components all by itself. The layout can be seen in Figure 10.

Now the EM setup was done the same way as for the piecewise simulation. The results are shown in Figures ?? and 11.

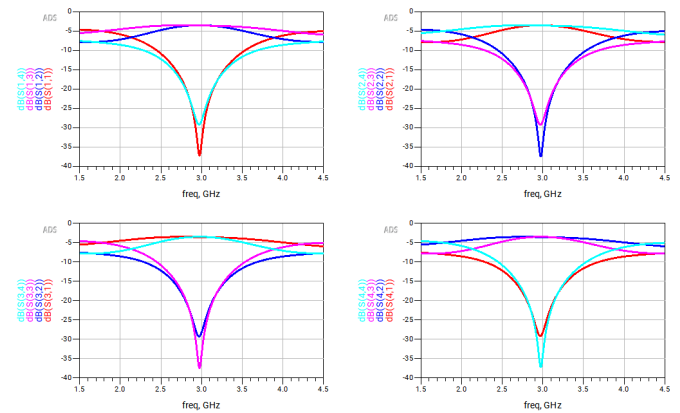


Fig. 11. EM simulation

The results show very low insertion loss between input and output, low reflection loss, and high isolation between ports 1 and 4.

Another point to note here is the phase difference between the signals received at ports 2 and 3. This can be found by taking the phase of S21/S31 and is supposed to be 90°. As observed in Figure 12 this has been achieved. Explanation for this lies in the lengths of the 35.35Ω and 50Ω 90° transmission lines' length. Both of them are of $\lambda/4$ length. This is 90° in electrical length. Which essentially makes the signal 90° out of phase. Since between input and Through ports only one $\lambda/4$ is there it makes the output at port to be 90° in phase. Since 2 such lines exist between input and coupled ports the output will be 180° out of phase. Difference between them has been observed in Figure 12.

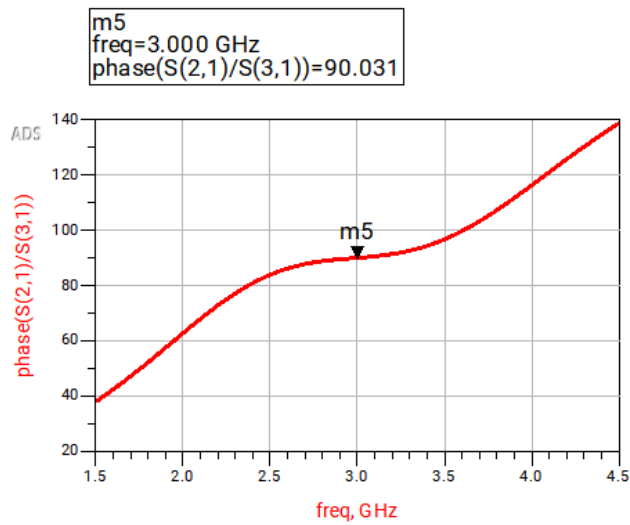


Fig. 12. Phase difference

III. CONCLUSION

The successful design and simulation of a 3 GHz, 50 Ω Branchline Coupler was achieved using Keysight ADS. The design process involved accurate impedance line modeling, S-parameter simulations, and electromagnetic simulations to verify performance. The final design exhibits excellent return loss, minimal insertion loss, and high isolation between the input and isolation ports. The resulting design meets the required specifications.

REFERENCES

- [1] D. M. Pozar, Microwave Engineering, 4th ed., Wiley, 2011.
- [2] Keysight Technologies, "Advanced Design System (ADS) User Manual," 2024
- [3] Keysight Technologies, "Advanced Design System (ADS) Distributed Components," February 2011